

SMART INDIA HACKATHON 2026

Strata Cadastre

3D ULPIN Generation and Vertical Property Mapping System

Final Project Report

Field	Detail
Problem Statement ID	26011
Problem Statement	3D ULPIN Generation and Vertical Property Mapping System
Ministry / Organisation	Ministry of Rural Development — Department of Land Resources
Theme	Smart Automation
Category	Software
Team Name	Strata_Cadastre_KP
Deployment region	Ahmedabad, Gujarat (pilot)

Contents

1. Executive Summary

Conventional land administration in India identifies parcels of ground. A land record is, in its underlying representation, a polygon on a plane. This was adequate when property was overwhelmingly horizontal, and it remains adequate for the majority of Indian parcels, which are agricultural or single-storey. It is no longer adequate for the vertical, layered cities that now hold a disproportionate share of the country's property value and virtually all of its urban infrastructure conflict.

A twelve-storey apartment building occupies one cadastral parcel and carries one Unique Land Parcel Identification Number. Twenty-four flats share that identifier. Basement parking bays, terrace and air rights, municipal water and sewer corridors running beneath the plot, and metro viaducts crossing the airspace above it have no unique spatial identity whatsoever. The consequence is not merely administrative untidiness: because these interests cannot be represented, conflicts between them cannot be detected at the point of registration. They surface later, as litigation, as excavation strikes on undocumented utility lines, and as alignment disputes that delay public infrastructure.

Strata Cadastre addresses this by extending the existing ULPIN into three dimensions. Every distinct property volume — apartment, basement bay, air-rights envelope, subsurface utility easement, elevated transport corridor — receives its own twenty-eight character identifier constructed on top of the statutory fourteen-digit ULPIN. Volumes are stored as true closed polyhedral geometry in a spatial database that rejects, at write time, any volume overlapping one already registered. Supporting this are an AI/ML pipeline that fuses drone imagery, LiDAR point clouds and sanctioned building plans; a geodetic layer that reconciles GNSS ellipsoidal heights with orthometric heights; and an interactive three-dimensional viewer with access tiering compliant with the Digital Personal Data Protection Act, 2023.

A working prototype exists and has been validated against analytically derived ground truth. The machine learning classifier achieves 99.39 per cent accuracy on held-out data. The volumetric deviation detector, which identifies unauthorised construction by computing the difference between as-built and sanctioned volumes, lands within 1.56 per cent of the mathematically correct answer. The database schema is verified by a nine-check automated suite. This report describes the system, the reasoning behind its design decisions, its measured performance, and — with equal emphasis — the boundaries of what has and has not been demonstrated.

2. Problem Definition and Background

2.1 The existing system

India's land records are managed under the Digital India Land Records Modernisation Programme, which assigns each surveyed parcel a fourteen-digit ULPIN. In urban Gujarat, property is recorded by the City Survey Office as a Property Card rather than the 7/12 extract used for agricultural land. Registration of transactions occurs at Sub-Registrar offices, while revenue administration, mutation and certain possession and easement disputes fall to the City Mamlatdar acting as Executive Magistrate under the Mamlatdar's Courts Act, 1906.

Each of these instruments describes a parcel by its boundary on the ground. None carries a field for elevation, floor level, or volumetric extent. This is not an oversight in implementation; it is a property of the data model.

2.2 What cannot be expressed

The following interests are routinely created, bought, sold, litigated and built upon in Indian cities, and none of them can be represented in a two-dimensional cadastre:

- **Individual apartment units.** A flat is a volume bounded on six sides. The parcel record describes only its shadow on the ground, shared with every other flat in the building.
- **Basement and parking volumes.** Among the most frequently disputed and least documented assets in Indian housing, often sold informally or bundled ambiguously with a flat.
- **Terrace and air rights.** Transferable Development Rights are actively traded in Gujarat and Maharashtra. A TDR certificate represents buildable volume, yet is tracked entirely separately from the space it refers to, because that space has no identifier.
- **Subsurface utility corridors.** Water mains, sewers and fibre routes are held as easements by utilities, not by the landowner. Where undocumented, they are struck during excavation.
- **Elevated transport corridors.** A metro viaduct crosses the airspace above private plots. The corporation owns the corridor volume; the landowner retains the land. Both facts are simultaneously true, and no single 2D record can hold them.

2.3 The consequence

The core failure. Because overlapping vertical interests cannot be represented, they cannot be detected. A registration system that has no geometry cannot refuse a transaction that conflicts with an existing one — it can only record the document and leave the conflict to be discovered later, in court or with an excavator.

It is worth being precise about the limits of this argument, because a well-prepared reviewer will test it. Flats are not wholly unidentified today: they carry sale deeds, society share certificates, municipal property tax tenement numbers and often RERA registration. These are genuine identifiers. What they are not is spatial. A tenement number cannot establish that a terrace extension on the fourth floor overhangs a balcony on the third, because it encodes no geometry at all. The argument for three-dimensional identity is therefore strongest precisely where no alternative identifier exists — subsurface and airspace interests — and weakest, though still valid, for apartments.

3. Proposed Solution

Strata Cadastre is a volumetric cadastre: a land administration system in which the unit of record is a closed three-dimensional volume rather than a planar polygon. It is designed as an extension of the existing system, not a replacement for it.

3.1 Design principles

- **Extend, never replace.** The statutory 2D ULPIN remains authoritative. Three-dimensional units are child records that reference it. No re-survey is required to begin, and rural and single-storey parcels are untouched.
- **Legal correctness before geometric correctness.** Indian apartment ownership is, in law, an Undivided Share in Land combined with an exclusive right to possess a spatial volume. The data model represents these as two linked records, following ISO 19152 (LADM), rather than flattening them into one row.

- **Detection at write time.** Topological validation occurs when a volume is registered, not when a dispute arises. An overlapping volume raises an exception and no record is created.
- **Fusion over single-sensor inference.** No single data source is trusted absolutely. Conflicts are resolved by an explicit precedence order, and every spatial node carries its own confidence and uncertainty.
- **Privacy by construction.** Access tiering under the DPDP Act, 2023 is implemented in the data model and enforced by database views, not applied as a filter at the presentation layer.

3.2 The 3D-ULPIN specification

The identifier is twenty-eight characters, composed of five segments. The fourteen characters at its centre are the existing government ULPIN, unmodified.

240124	11000088421000	T	906	T010	
				+-	Unit ID (4)
				+-----	Floor index (3)
				+-----	Vertical type (1)
				+-----	Existing 2D ULPIN (14)
				+-----	LGD locator (6)

Six vertical types are defined:

Code	Type	Example interest
B	Basement / subsurface	Parking bay beneath the plinth
G	Ground	Retail unit at street level
F	Floor	Residential apartment on an upper storey
T	Terrace / air-rights	Unbuilt volume above the sanctioned roof, tradeable as TDR
U	Subsurface utility	Municipal water or sewer easement
E	Elevated corridor	Metro viaduct crossing plot airspace

The floor index is stored unsigned within the identifier because its sign is implied by the vertical type; the database retains a signed floor index separately for querying. Identifier format is enforced by a regular-expression CHECK constraint in the schema, so a malformed identifier cannot be persisted.

4. System Architecture

The system is organised as a six-stage pipeline feeding a spatial database, an API layer and a three-dimensional client.

```

Drone imagery · LiDAR · CAD/IFC · GNSS-CORS · DEM/DSM
      |
[1] GEODETIC NORMALISATION      H = h - N
      |
[2] AI/ML FUSION                footprint extraction, nDSM,
                                ICP alignment, precedence
      |
[3] TOPOLOGY VALIDATION        zero-overlap enforcement
                                dV = as-built \ sanctioned

```

[4] 3D-ULPIN ASSIGNMENT	28-character identifier
[5] PostGIS 3D + SFCGAL	-> [6] FastAPI -> 3D viewer

4.1 Geodetic layer

GNSS and CORS receivers report ellipsoidal height, measured against the WGS84 reference ellipsoid. Cadastral records and construction drawings are referenced to orthometric height — height above mean sea level. The two differ by the geoid undulation, and over India that difference is large: the subcontinent sits on the northern shoulder of the Indian Ocean Geoid Low, the most negative geoid anomaly on Earth. Undulation runs from roughly -45 metres near the Himalayan foothills to near -100 metres at the southern tip. Ignoring the correction would introduce errors far larger than any building.

The module implements $H = h - N$, and emits two distinct elevation fields for every spatial node: an absolute geodetic elevation referenced to mean sea level, and a relative plinth elevation measured from the building's own ground slab. The first is comparable across jurisdictions; the second is what site engineers and the deviation detector actually reason about, and is unaffected by geoid model error.

4.2 Fusion engine

The fusion engine extracts building footprints from drone orthomosaics, applying a relief-displacement correction that maps the photographed rooftop polygon back to its true ground-nadir position. It derives a normalised height model by subtracting a terrain model from a surface model, and aligns the resulting envelope to the sanctioned CAD plan using Iterative Closest Point registration.

A deliberate constraint. Floor count is never inferred from height alone. Double-height lobbies and mezzanines make naive division by an assumed floor height unreliable. The LiDAR-derived apex height is reconciled against the count declared in the sanctioned plan, and a discrepancy is flagged for review rather than silently resolved in favour of either source.

Where sources disagree on a boundary, resolution follows a strict precedence order, with each node retaining a confidence score and a positional uncertainty in centimetres:

CORS-RTK GNSS (± 10 cm) > LiDAR point cloud > sanctioned CAD > legacy 2D GIS

4.3 Database design

The schema targets PostgreSQL with PostGIS and the SFCGAL backend. Seven tables implement the model: a base 2D parcel table, a vertical 3D unit table storing PolyhedralSurfaceZ geometry, an append-only cadastral lifecycle log implementing a state machine, a volumetric property tax registry, an encumbrance registry, an owner registry, and a point-cloud reference table.

Two of these were not named in the original specification but proved necessary. The owner registry exists because the access-tiering requirement names owner KYC and sale deed hashes as registrar-tier data without providing anywhere to hold them; it stores verification status and document hashes, never raw identity numbers. The point-cloud reference table enforces the requirement that raw LiDAR never enters the relational database: it holds only pointers to externally hosted COPC or EPT endpoints.

Topological validation

A trigger fires before every insert or geometry update on the vertical unit table. It queries all existing units on the same floor of the same parcel, filters candidates with a bounding-box-accelerated three-dimensional intersection test, and for each candidate computes the actual volume of intersection. Only an intersection volume above a small floating-point tolerance constitutes a conflict.

This two-stage structure is essential rather than an optimisation. Adjacent apartments sharing a party wall will always register as intersecting under a boolean test; treating that as a conflict would reject every legitimate neighbouring unit. Distinguishing a shared face from a genuine shared volume is the difference between a working validator and one that rejects the entire building.

5. Machine Learning Approach

The problem statement calls for AI/ML capabilities in automated building extraction, floor segmentation, vertical parcel delineation and topology validation. Of these, the one that genuinely requires a learned model is point cloud classification.

5.1 Why a learned model is necessary here

A simple heuristic can separate low points from high points, and this is sufficient to find the ground. It cannot separate a tree from a building, because both are high. That distinction matters materially: vegetation misclassified as building inflates the extracted footprint, which inflates every volume, every ULPIN boundary and every tax assessment derived from it. Shape, not elevation, is what separates them, and shape is what a learned model over geometric features captures.

5.2 Features and model

Eight geometric features are computed for each point from its local neighbourhood:

Feature	What it captures
Height above cell minimum	Local elevation above the lowest point in the grid cell
Height above neighbourhood minimum	Elevation above the lowest point within a radius
Planarity	$(\lambda_2 - \lambda_3) / \lambda_1$ — high on roofs, walls and ground
Linearity	$(\lambda_1 - \lambda_2) / \lambda_1$ — high on edges and wires
Sphericity	λ_3 / λ_1 — high on scattered volumes such as foliage
Verticality	Derived from the surface normal; high on walls
Neighbourhood density	Point count within a fixed radius
Local height standard deviation	Vertical spread, a strong vegetation cue

A Random Forest classifier is trained over these features. The choice is deliberate rather than a compromise: with a training set of this size and eight well-engineered features, a forest outperforms a small neural network

and trains in seconds on a laptop. Point-cloud deep learning architectures require volumes of labelled data that a hackathon cannot produce, and an undertrained network is worse than a well-chosen classical model.

5.3 Measured results

Metric	Result
Held-out classification accuracy	99.39%
Building extraction precision	98.66%
Building extraction recall	99.58%
Vegetation misclassified as building	0.94%
Most discriminative feature	Height above neighbourhood minimum (0.257)
Second most discriminative	Sphericity (0.230)

That sphericity ranks second is a useful confirmation that the model learned what it was intended to learn: foliage is geometrically scattered while walls and roofs are planar, and the model is discriminating on that basis rather than on height alone.

5.4 Floor plan segmentation

Boundary and room extraction from raster floor plans uses classical computer vision: Otsu thresholding, morphological cleanup, contour detection and Douglas-Peucker simplification for the outer boundary; distance transform and watershed for the rooms.

The watershed step replaced an initial contour-hierarchy approach that failed on the first realistic test. Treating rooms as holes inside the outer wall contour works only when every room is fully enclosed. Real floor plans have doorways, so the interior is one connected region and hierarchy returns a single room. Morphological closing cannot bridge a door gap without destroying the wall geometry being measured. The distance transform peaks at room centres and dips at doorways, so thresholding it yields one seed per room and watershed places each boundary at the doorway — which is where a surveyor would place it.

Extracted boundary area falls within 1.32 per cent of the true dimensions on a test plan of known size, and all four rooms are correctly segmented despite door gaps.

6. Validation and Testing

Every module in the system is independently executable and validates its own output against analytically derived ground truth rather than against expectation. The following figures are produced by running the code, not estimated.

Component	Validation	Result
Geodetic conversion	Round-trip $h \rightarrow H \rightarrow h$	Exact to 1e-9 m
Relief displacement	Forward model then inverse	0.000000 m error

Component	Validation	Result
ICP alignment	Recovery of a known 12° transform	0.022° / 1 mm
Plinth extraction	Against known synthetic plinth	±1 cm
Apex extraction	Against known building height	±7 cm
Deviation detector	Against analytic volume	1.56% error
ML classifier	Held-out test split	99.39% accuracy
Floor plan boundary	Against known plan dimensions	1.32% area error
LAS reader	Write / read round trip, 5,000 points	Sub-millimetre
Database schema	Nine-assertion suite	9 / 9 pass

6.1 A defect found by the verification suite

Units error in volume computation. Geometry is stored in EPSG:4326, whose X and Y are degrees while Z is metres. The volume trigger applied ST_Volume directly to that geometry, producing a figure in degree-squared-metres — wrong by approximately ten orders of magnitude. It never raised an error and never returned null.

This is the most instructive result in the project. A property tax assessment derived from that column would have been wrong by a factor of roughly eleven billion, with nothing visibly broken anywhere in the system. It was found not by inspection but by writing an assertion that the computed volume of a known $15 \times 11 \times 3.2$ metre unit should be approximately 528 cubic metres. The fix transforms geometry into a projected coordinate system before measurement, and the assertion now guards against regression permanently.

The general lesson generalises beyond this project: defects that produce no error are more dangerous than those that crash, and only assertions against independently known values will catch them.

7. Feasibility Analysis

7.1 Technical feasibility

The system is built entirely on mature open standards — ISO 19152, OGC CityGML 3.0, OGC 3D Tiles, COPC and EPT — with no unproven technology anywhere in the stack. A working prototype runs the full pipeline end to end. Deployment is self-hosted Docker on commodity hardware, which matters because government land records run in NIC and state data centres rather than on external cloud platforms.

7.2 Economic feasibility

Because the system extends the existing ULPIN rather than replacing it, adoption begins from records already digitised under DILRMP; no re-survey is required to start. Volumetric assessment permits municipal property tax to be levied on actual cubic metres and usage type, broadening the base without raising rates. Automated deviation detection surfaces regularisation cases that are currently found only by complaint, and reduces manual survey cost. Infrastructure cost is low: the entire stack runs on a single modest server.

7.3 Legal feasibility

The data model follows Indian apartment ownership law as it actually operates, representing the Undivided Share in Land and the exclusive right to a spatial volume as two linked records. Access tiering implements the Digital Personal Data Protection Act, 2023 at the database level. Because the statutory 2D ULPIN remains authoritative and unmodified, the system introduces no conflict with existing records.

The binding constraint is legal, not technical. A Sub-Registrar registers documents; he does not verify title. A perfectly accurate volumetric record changes nothing unless some instrument requires it to be consulted before registration. That is legislative work, not software work — and it is the honest answer to the question of what would stop this system from mattering.

International precedent is instructive and cuts slightly against an engineering-first approach. Jurisdictions that have implemented volumetric cadastre — the Netherlands, Sweden, and Queensland among them — generally legislated strata or volumetric title first and built the spatial infrastructure afterwards. It is more honest to present this system as the infrastructure that becomes necessary once such a framework exists than to claim it forces the framework into being.

7.4 Social feasibility

For flat buyers, the system provides a verifiable spatial identity, RERA status and encumbrance check before purchase, and gives individual title to parking, basement and terrace rights that are today among the least documented assets in Indian housing. For registration offices, conflicting claims become impossible to register rather than arguable in court. For utilities and infrastructure agencies, registered subsurface and airspace rights prevent the excavation strikes and alignment disputes that delay public works.

8. Risks and Mitigation

Risk	Mitigation
Sub-metre LiDAR is not publicly available for Ahmedabad. National DEMs are 30 m resolution — a 15 m building is half a pixel.	Phone photogrammetry: 100–200 overlapping photographs processed in Meshroom or COLMAP produce a genuine point cloud. No drone permit, no cost, legal anywhere.
Legacy 2D records carry metre-scale positional error that would propagate into 3D.	Precedence matrix with per-node confidence and uncertainty. Legacy data is flagged rather than silently overwritten.
SFCGAL is unavailable on some hosts. AWS RDS does not offer it; Alpine PostGIS images cannot run it.	Verified deployment path: Neon documents support, and Debian PostGIS images work. The nine-check suite proves any candidate database before it is relied upon.
City-scale point cloud volume exceeds what a relational database can hold.	COPC and EPT streaming keeps raw clouds in object storage; only vectorised volumes are persisted.
Municipal adoption requires process change inside registration offices, not only software.	Phased rollout targeting dense urban parcels first; rural and single-storey records remain untouched.
Drone operation is restricted near Ahmedabad airport under DGCA rules.	Ground-based photogrammetry requires no airspace clearance; drone capture is optional rather than necessary.

9. Primary Field Research

Requirements were checked against practice rather than assumed. Three lines of enquiry were pursued in Ahmedabad:

- **City Mamlatdar and Executive Magistrate.** Possession and easement dispute process under the Mamlatdar's Courts Act, 1906, including which records are consulted when two parties claim the same space, and whether those records represent anything above or below ground level.
- **City Survey Office.** The Property Card format that governs urban property in Gujarat, and specifically which fields exist and which vertical attributes have nowhere to be recorded. This enumeration is the empirical basis of the gap analysis.
- **Local developer.** Sanctioned plan structure, how Undivided Share is calculated and allotted per flat, the building-permission and occupancy-certificate workflow, and how setback or FSI deviations are currently detected in practice.

All material collected was anonymised at source. No owner identity, no unredacted deed and no personal data was retained — a discipline consistent with the DPDP posture the system itself implements.

10. Limitations and Honest Assessment

A system whose boundaries are known is worth more than one that overclaims. The following limitations are stated without qualification.

- **The machine learning model is trained on synthetic point clouds.** The model, the feature engineering and the reported metrics are real and reproducible; the training data is simulated, because free labelled LiDAR covering Ahmedabad does not exist in the public domain. Retraining on classified tiles is the immediate next step, and field accuracy will be lower than the held-out figures reported here.
- **The API currently persists to SQLite.** The PostGIS schema is real and independently verified, but connecting the API to it is a change to two functions that has not yet been made.
- **DXF plan reading is untested.** The library required was unavailable in the development environment. The function is marked as unverified in the source rather than presented as working.
- **No real building has been processed end to end.** Field research to source a genuine sanctioned plan and a photogrammetric point cloud is underway but not complete.
- **The viewer renders entities, not 3D Tiles.** This is correct at single-parcel scale. True city-scale rendering requires a tiling pipeline.
- **The geoid model is a coarse regional approximation.** A precomputed lookup table interpolated over India, accurate to roughly $\pm 1\text{--}3$ metres, stands in for full EGM2008 synthesis. It is adequate to drive the pipeline and verify plausibility; it is not survey-grade and must not underpin a legal boundary.

11. Future Work

- **Retrain the classifier on field-labelled LiDAR.** Publicly available classified tiles provide a path to defensible field accuracy figures.
- **Connect the API to PostGIS.** Replace the SQLite persistence layer, bringing the verified schema and its topological triggers onto the live request path.

- **Process a real Ahmedabad building.** A sanctioned plan from a developer combined with a phone-photogrammetry point cloud would convert the deviation detector from a demonstration into a case study.
- **Implement 3D Tiles streaming.** Required for district-scale rendering performance.
- **Integrate a survey-grade geoid model.** Either a full EGM2008 grid or, where mandated, the Survey of India model.
- **CityGML / CityJSON export.** GeoJSON export exists; full LOD2 volumetric interchange is the next interoperability milestone.
- **Engage on the legal framework.** The system's value is contingent on volumetric title being recognised. Policy engagement is as important as further engineering.

12. Conclusion

Strata Cadastre demonstrates that volumetric property identity can be generated, validated and served using mature open technology, and that doing so requires no disruption to the existing two-dimensional record. The twenty-eight character identifier extends the statutory ULPIN rather than competing with it. Topological validation converts a class of ownership conflict from something litigated after the fact into something that cannot be registered in the first place. The deviation detector finds unauthorised construction as a by-product of the same computation that assigns identifiers.

The measured results — 99.39 per cent classification accuracy, deviation detection within 1.56 per cent of ground truth, a nine-check verified database schema — establish that the approach works on the data it has been given. The limitations, equally measured, establish what remains to be proven: principally that it works on real survey data from a real Indian building, and that a legal framework exists to give the resulting records force.

The most useful thing learned in building this system was not a technique but a discipline. A verification suite written against independently known values found a defect in our own schema that produced numbers wrong by ten orders of magnitude while raising no error at all. In land administration, where a record may stand for decades and a family's principal asset depends on it, silent wrongness is the failure mode that matters. Designing for its detection is not an optional refinement.

13. References

Standards

- ISO 19152:2012 — Land Administration Domain Model (LADM). International Organization for Standardization.
- OGC CityGML 3.0 Conceptual Model Standard. Open Geospatial Consortium.
- OGC 3D Tiles Specification. Open Geospatial Consortium.
- Cloud Optimized Point Cloud (COPC) specification. copc.io
- OGC API — Features. Open Geospatial Consortium.

Government sources

- Digital India Land Records Modernisation Programme (DILRMP), Department of Land Resources, Ministry of Rural Development.
- ULPIN — Unique Land Parcel Identification Number. Department of Land Resources.
- SVAMITVA Scheme — drone-based survey of inhabited rural areas.
- The Digital Personal Data Protection Act, 2023. Government of India.
- AnyRoR — Gujarat land record portal, Revenue Department, Government of Gujarat.
- The Mamlatdar's Courts Act, 1906.
- Gujarat Ownership Flats Act — statutory basis for Undivided Share in Land.

Technology

- PostGIS and SFCGAL documentation — three-dimensional spatial operations.
- PDAL — Point Data Abstraction Library. GDAL — Geospatial Data Abstraction Library.
- scikit-learn — Random Forest classifier.
- CesiumJS, Three.js, FastAPI, OpenCV.

Appendix A — Repository Contents

File	Purpose
geodesy.py	Ellipsoidal to orthometric height conversion
pipeline.py	Fusion engine, topology validation, deviation detection
ml_classifier.py	Random Forest point cloud classifier
floorplan_extract.py	Floor plan boundary and room segmentation
ingest_real.py	LAS / PLY / XYZ readers and georeferencing
geo_io.py	GeoJSON parcels, DEM/DSM rasters, 3D export
main.py	FastAPI endpoints and the 3D-ULPIN generator
db/schema_3d_cadastre.sql	PostGIS 3D schema with triggers and RBAC views
db/schema_patch_01.sql	Units and formatting corrections
db/verify_schema.sql	Nine-assertion database verification suite
web/demo.html	Standalone 3D viewer — no install required
web/CadastreViewer3D.jsx	React and CesiumJS viewer component

Appendix B — Reproducing the Results

Every figure in this report can be reproduced by running the corresponding module:

```

pip install -r requirements.txt

python geodesy.py          # height conversion, round-trip check
python pipeline.py        # fusion, ICP, deviation detection
python ml_classifier.py    # trains classifier, prints held-out metrics
python floorplan_extract.py # plan extraction vs known dimensions
python ingest_real.py      # LAS/PLY round trip, scale correction
python geo_io.py          # GeoJSON and DEM round trip
python main.py            # end-to-end ingest -> ULPIN -> topology

docker compose up -d
docker compose exec -T db psql -U strata -d strata \
  -v ON_ERROR_STOP=1 < db/verify_schema.sql

```